

## Using concepts of stochastic processes to analyse financial markets

In mathematics, a stochastic process can be thought of as a continuous function, however unlike smooth functions such as polynomials, trigonometric, or exponential functions, subsequent values follow a random distribution. This means that there is no formula for determining what the value of  $f(t)$  will be at some time  $t$ ; an alternative way of thinking about a stochastic process is as a sequence of random outcomes that are infinitesimally close together such that they form a continuous yet erratic function. If the random distribution of the increments of a stochastic process is normally distributed, then the process is called a Brownian motion, also known as a Wiener process – this is the process that this essay will focus on. A standard Brownian motion, denoted as  $W_t$  has these following properties:

$$\begin{aligned} W_0 &= 0 \\ W_t &\sim N(0, t) \Rightarrow \sigma^2 \propto t \\ \mathbb{E}(W_t) &= 0 \end{aligned}$$

This shows that Brownian motion is normally distributed with a mean, or expected value, of 0, and a variance proportional to the time given. In other terms, the variance, or spread of where the value is likely to be, increases over time, as using this definition, we do not need to begin with a certain time to determine its value:  $t = 5$  could be immediately calculated before a time such that  $0 \leq t < 5$ . Brownian motion can also be defined in increments:

$$W_t - W_s \sim N(0, t - s) \quad 0 \leq s < t$$

Here,  $t$  denotes the time that the value,  $W_t$ , will be calculated from, and  $s$  denotes a previous time; crucially, since  $s < t$ , we know the value of  $W_s$ . Similarly to Brownian motion without being calculated in increments, the following properties hold:

$$\mathbb{E}(W_t - W_s) = 0$$

The expected value is still equal to zero and the variance is equal to the difference in times. In fact, these two definitions describe the exact same process which can be shown if we let  $s = 0$  and  $t > 0$ :

$$W_t - W_s \sim N(0, t - 0)$$

$$W_0 = 0, s = 0 \Rightarrow W_s = 0$$

$$W_t \sim N(0, t)$$

This is the exact same definition of Brownian motion without increments. Another key property of this process is that each increment is independent such that for any choice of  $t_1 < t_2 < t_3 < \dots < t_n$  the random values of each increment  $W_{t(2)} - W_{t(1)}$ ,  $W_{t(3)} - W_{t(2)}$ , ...,  $W_{t(n)} - W_{t(n-1)}$  are jointly independent; this implies that Brownian motion is a Markov process: the past is irrelevant in determining the probability of future values. Despite being continuous everywhere, the derivative of this process does not exist – this becomes clear when we view a graph of Brownian motion as it is fractal-like and infinitely ‘jagged’, meaning no matter how far you zoom into a section of the graph, it will never become smoother like other differentiable functions. Brownian motion, and in fact all stochastic processes, are non-differentiable, which is a key difference between a

stochastic process and a smooth, non-random function. This fact can be shown if we attempt to use first principles of differentiation:

$$W'_t = \lim_{h \rightarrow 0} \frac{W_{t+h} - W_t}{h}$$

Recall that the increments of Brownian motion follow a random normal distribution:

$$W_t - W_s \sim N(0, t - s) \quad 0 \leq s < t$$

$$X = W_t - W_s$$

Where the random variable  $X$  is equal to  $W_t - W_s$ . This can be alternatively written as:

$$W_{t+\Delta t} - W_t \sim N(0, \Delta t) \quad \Delta t > 0$$

This means that the numerator of the limit derived from first principles of differentiation follows a normal distribution with a mean of 0 and variance of  $h$ . Since a normal distribution can be standardized using the following:

$$\text{if } X \sim N(\mu, \sigma) \text{ then } Z = \frac{X - \mu}{\sigma} \text{ where } Z \sim N(0, 1)$$

The random value of  $W_t - W_s$  can be written as:

$$Z = \frac{W_t - W_s}{\sqrt{h}} \Rightarrow W_t - W_s = Z\sqrt{h}$$

Therefor the limit becomes:

$$W'_t = \lim_{h \rightarrow 0} \frac{Z\sqrt{h}}{h} = \lim_{h \rightarrow 0} \frac{Z}{\sqrt{h}}$$

This limit diverges to infinity as  $h$  approaches 0, therefor the derivative of the original function, the Brownian motion, is undefined and therefor does not exist in this standard sense. Crucially, no matter what the value of the random standardized variable  $Z$  is, the limit still diverges to infinity. An important property of Brownian motion is that it is continuous; a continuous function is such that a small change in the input causes a small variation in the value of the function, meaning there are no sudden changes in value and there are no holes or asymptotes. A more rigorous definition is as follows:

$$\int_b^a f'(x)dx = f(a) - f(b), \quad a > b$$

Or, for any function whether it is differentiable or not:

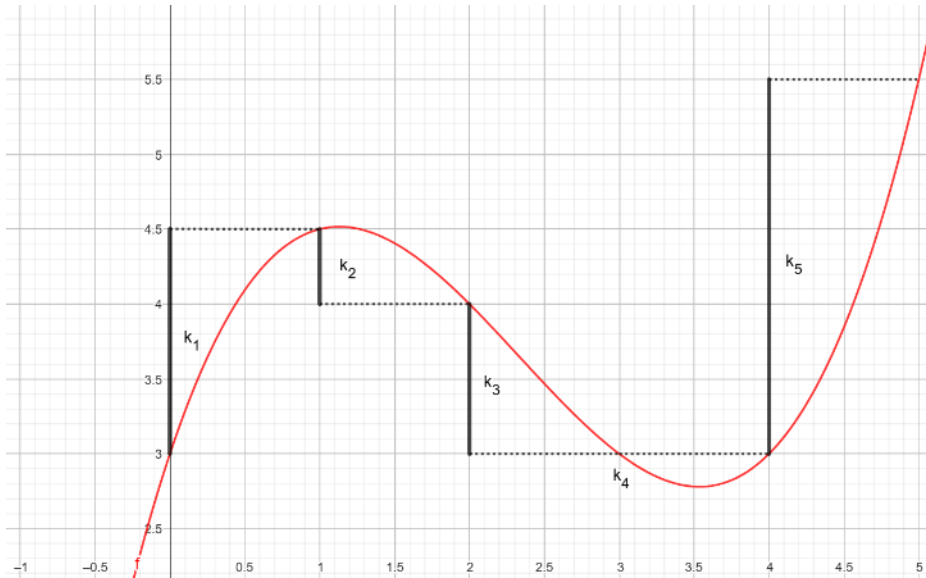
$$\forall \varepsilon > 0: \exists \delta > 0 \text{ such that } |x - x_0| < \delta$$

$$\text{This should imply that } |f(x) - f(x_0)| < \varepsilon$$

Meaning for any epsilon chosen, no matter how small, the distance between some  $f(x)$  and a fixed  $f(x_0)$  must be less than epsilon, given that the distance between  $x$  and  $x_0$  is less than delta. This breaks in non-continuous domains, as a vertical asymptote contains an undefined value, there is no epsilon greater than 0 equal to the difference of some defined  $f(x)$  and an undefined value.

Another essential property called the variation of a function is a key element that distinguishes differentiable functions from stochastic processes. Variation can be defined as how much a function changes over an interval across its domain. This can be calculated by dividing the interval into a certain number of equal partitions, where the total variation across the interval is the sum of the absolute value of the change in height of each partition. This measure is called first order variation (or total variation). The following example demonstrates the total variation of a simple cubic equation using 5 partitions:

$$\text{total variation} = V_b^a(f) = \lim_{n \rightarrow \infty} \sum_{k=0}^n \left| f\left(b + \frac{(k+1)(a-b)}{n}\right) - f\left(b + \frac{k(a-b)}{n}\right) \right|, \quad a > b$$



$$f(x) = \frac{1}{4}(x-2)^3 - \frac{1}{4}x^2 + 5$$

$$k_1 = f(1) - f(0), k_2 = f(2) - f(1), \dots, k_n = f(n) - f(n-1)$$

$$V_0^5(f) \approx \sum_{k=0}^5 \left| f\left(\frac{5(k+1)}{5}\right) - f\left(\frac{5k}{5}\right) \right| = \sum_{k=0}^5 |f(k+1) - f(k)|$$

$$V_0^5(f) \approx k_1 + k_2 + k_3 + k_4 + k_5$$

$$V_0^5(f) \approx \frac{3}{2} + \frac{1}{2} + 1 + 0 + \frac{5}{2} = \frac{11}{2}$$

In this example, the first order variation is approximately 5.5 if we use 5 partitions. This number becomes more accurate as the number of partitions increase. If a graph rapidly changes during a certain interval, its first order variation will be much greater in that interval than the first order variation of a function that is flatter. If a function is differentiable, its total variation will always be finite no matter what the given interval is, meaning it has a bounded variation. On the contrary, a stochastic process such as Brownian motion will have infinite first order variation across any given interval, and can be described as having an unbounded variation. Nevertheless, this does not mean the total change of a stochastic process cannot be determined – a different type of variation, quadratic variation, is finite for these types of processes. Quadratic variation, or second order variation, is defined in exactly the same way as total variation, however the differences in values of the partitions are squared:

$$X_b^a(f) = \lim_{n \rightarrow \infty} \sum_{k=0}^n \left( f\left(b + \frac{(k+1)(a-b)}{n}\right) - f\left(b + \frac{k(a-b)}{n}\right) \right)^2, \quad a > b$$

In fact, this concept can be generalised to third order or beyond, where the difference in height of partitions is raised to the order, for example third order variation is the sum of the cubed differences in partitions:

$$V_b^a(f) = \lim_{n \rightarrow \infty} \sum_{k=0}^n \left( f\left(b + \frac{(k+1)(a-b)}{n}\right) - f\left(b + \frac{k(a-b)}{n}\right) \right)^3$$

Going further, a completely different function can be applied to the difference in heights of subsequent partitions:

$$\phi_b^a(f) = \lim_{n \rightarrow \infty} \sum_{k=0}^n \phi \left( f\left(b + \frac{(k+1)(a-b)}{n}\right) - f\left(b + \frac{k(a-b)}{n}\right) \right)$$

For example, the function  $\phi(x)$  could be  $\ln(x)$ , where the differences would be fed into the function before being summed up. For quadratic variation the function  $\phi(x)$  in this case is simply  $f(x) = x^2$ . We calculate quadratic variation in the interval  $[0, t]$  since  $W_0$  is defined to be 0 in Brownian motion, so the lower bound denoted by  $b$  in the above formula becomes 0:

$$X_a(f) = \lim_{n \rightarrow \infty} \sum_{k=0}^n \left( f\left(\frac{a(k+1)}{n}\right) - f\left(\frac{ak}{n}\right) \right)^2, \quad a > 0$$

It is common to have the variable time denoted by  $t$ , and let  $P$  range over partitions of the interval  $[0, t]$ :

$$X_t = \lim_{\|P\| \rightarrow 0} \sum_{k=1}^n (X_{t_k} - X_{t_{k-1}})^2$$

Consequently, the quadratic variation will be finite for a stochastic process, yet quadratic variation for a differentiable function will converge to zero as the number of partitions approaches infinity. A remarkable aspect of Brownian motion is that its quadratic variation is not random, despite being defined as following a random normal distribution in each increment. In fact, incredibly, the quadratic variation in this case converges to the time, seemingly creating a predictable value from a completely random process. In order to prove this result, the following fact from probability theory will be used:

$$\sigma^2 = \mathbb{E}[(X - \mu)^2]$$

Where  $\sigma^2$  is the variance, which is equal to the expected value of the square of the distance from a random variable under a normal distribution to the mean. Since by definition of Brownian motion:

$$W_t - W_s \sim N(0, t - s) \quad 0 \leq s < t$$

Which can be rewritten as

$$W_{t_{i+1}} - W_{t_i} \sim N(0, t_{i+1} - t_i)$$

Where  $t_i$  is a partition in the interval  $[0, t]$ , then:

$$\text{Var}(W_t - W_s) = \mathbb{E}[(W_t - W_s)^2] = t_{i+1} - t_i$$

Recall that the quadratic variation of a stochastic process is:

$$Q = \lim_{n \rightarrow \infty} \sum_{i=1}^n (W_{t_{i+1}} - W_{t_i})^2$$

Taking expectations of the whole sum gives:

$$\mathbb{E}(Q) = \mathbb{E} \left( \lim_{n \rightarrow \infty} \sum_{i=1}^n (W_{t_{i+1}} - W_{t_i})^2 \right)$$

Since the sum is a linear operator, we can take the sum outside:

$$\mathbb{E}(Q) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mathbb{E}((W_{t_{i+1}} - W_{t_i})^2)$$

$$\text{recall that } \mathbb{E}[(W_t - W_s)^2] = t_{i+1} - t_i$$

$$\text{so } \mathbb{E}(Q) = \lim_{n \rightarrow \infty} \sum_{i=1}^n t_{i+1} - t_i$$

Since the increments in the interval  $[0, t]$  are all equally spaced apart:

$$\text{for } i: i \leq n \quad t_{i+1} - t_i = \Delta t = \frac{t}{n}$$

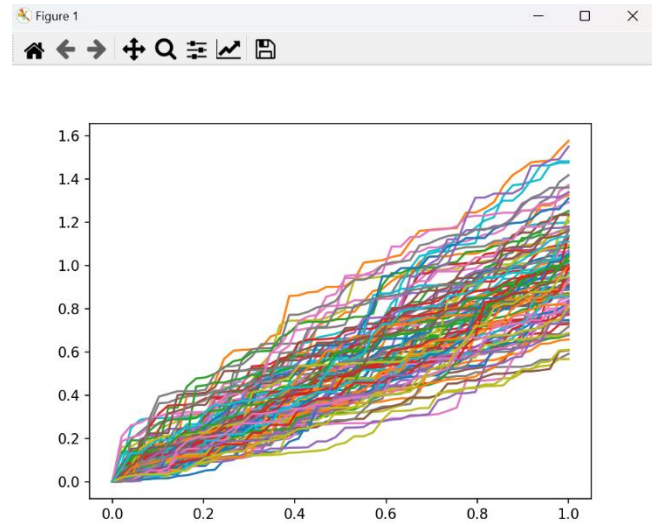
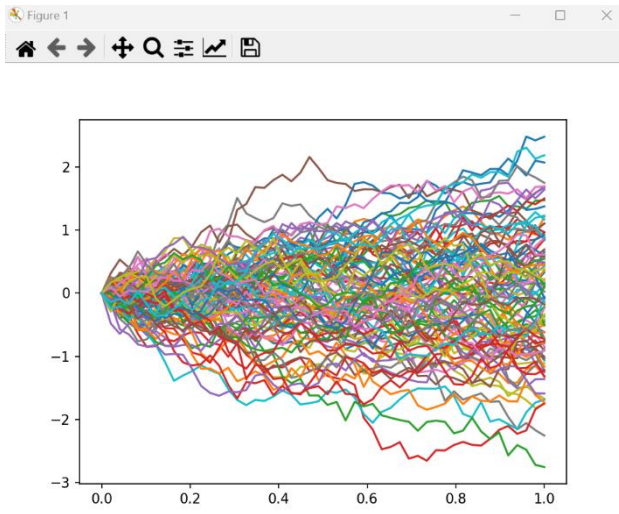
$$\text{so } \mathbb{E}(Q) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{t}{n} = n \times \frac{t}{n}$$

$$\therefore \mathbb{E}(Q) = t$$

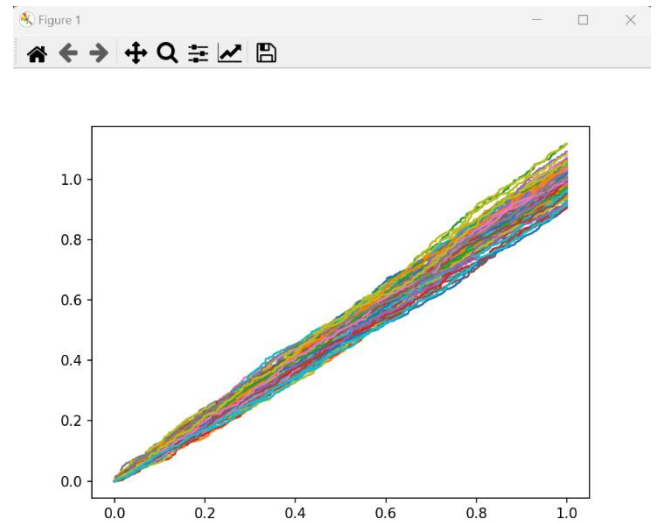
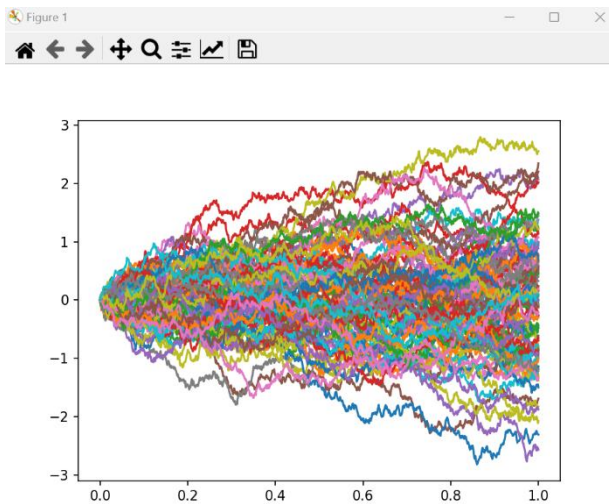
The expected value of the quadratic variation of Brownian motion in the interval  $[0, t]$ , where  $t$  is any number greater than 0, turns out to equal  $t$ . Since Brownian motion is infinitely 'jagged', we take the limit of the number of partitions as they approach infinity, meaning the real value converges to  $t$ .

This result is remarkable as despite the process being non-differentiable and defined to be randomly normally distributed, the quadratic variation is entirely predictable – this can have real applications such as if a financial market follows an approximate Brownian motion pattern. Amazingly, the fact that a stock market does not have an infinite number of random points strung together, but rather approximately follows a Brownian motion, enables us to predict with high probabilities whether the market will change drastically or not. To demonstrate this, the diagrams on the left, generated from Python, below show one hundred Brownian motions with varying number of points that make up the lines, denoted by  $n$ , and another graph on the right to show each of the one hundred motions' quadratic variation up to each one of the partitions:

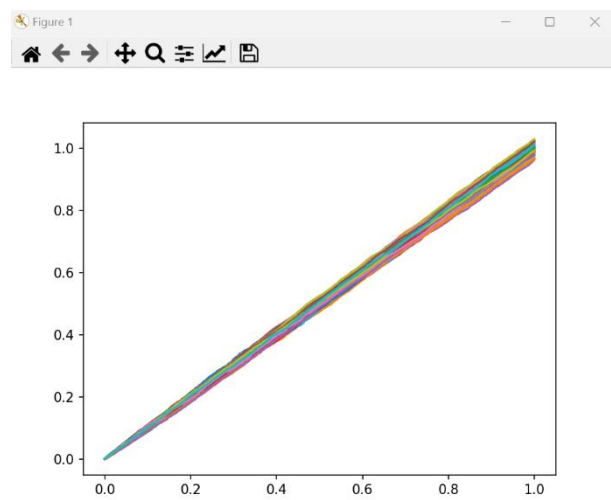
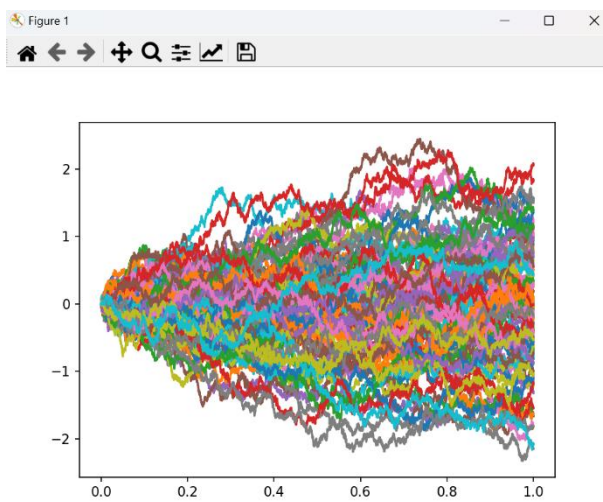
$n = 50$



$n = 1000$



$n = 10000$



As shown, the greater the number of points that make up the Brownian motions, the closer the quadratic variation graph becomes to the line  $y = x$ . As the number of points,  $n$ , approaches infinity, the graph of quadratic variation will converge exactly to the line  $y = x$ . As we know that it converges to this line, if we use a finite number of points for a Brownian motion, an individual motion must stay approximately fixed to the line, meaning if we see it is moving far away from the line, the probability of the Brownian motion returning to the line becomes greater after every increment. Similarly, if the quadratic variation of the motion is already close to the line  $y = x$ , then there is a greater chance that the motion will stay close to it rather than changing abruptly. Applying this concept, if a financial market can be modelled as an approximate Brownian motion, then the quadratic variation up to a certain time can be calculated – from this we can only analyse what the volatility of the market will be, not whether values will increase or decrease. This is because when calculating quadratic variation, since the differences in height of subsequent partitions is squared, quadratic variation is always positive.

Another clear pattern emerges from the diagrams of Brownian motion – nearly all of the motions seem to be contained within a parabolic shape. In fact, it can be shown that all Brownian motions that lie within one standard deviation away from the mean fall within the bounds of the square root of  $t$ . By definition of Brownian motion:

$$W_t \sim N(0, t)$$
$$\therefore \sigma^2 = t \Rightarrow \sigma = \sqrt{t}$$

This means that the variance of  $W_t$  is equal to  $t$ . Since the standard deviation is equal to the square root of the variance, one standard deviation of  $W_t$  equals the square root of  $t$ , meaning roughly 68% of Brownian motions will stay within this region.

In conclusion, using concepts of stochastic processes such as Brownian motion can be very useful in analysing seemingly random functions despite appearing to be completely unpredictable. Because stochastic processes are proven to be non-differentiable, standard integration and differentiation techniques cannot be used to analyse these functions, yet first or second order variation can give insights into how these random processes behave.

## References

- Stepbil, *Brownian Motion #1 (basic properties)* (2011), accessed 15 August 2024, [Brownian motion #1 \(basic properties\) \(youtube.com\)](#)
- Siegrist, K, *Standard Brownian Motion* (2022), accessed 15 August 2024, [18.1: Standard Brownian Motion - Statistics LibreTexts](#)
- Wikipedia: *Quadratic Variation* (2024), accessed 15 August 2024, [Quadratic variation - Wikipedia](#)
- Dai, P.T, supervised by Markowsky, G and Das, K (2021) *Properties of Brownian Motion*, Australian Mathematical Sciences Institute, accessed 17 August 2024, [dai\\_pu\\_ti\\_vrs-report.pdf \(amsi.org.au\)](#)
- mCoding, *Simulating Brownian Motion in Python* (2021), accessed 18 August 2024, [Simulating Brownian Motion in Python \(youtube.com\)](#)
- GeoGebra, accessed 18 August 2024, [Calculator Suite - GeoGebra](#)
- *Microsoft Visual Studio 2022*, accessed 18 August 2024 (desktop app)